

Gradient SNR and Loss-Curve Oscillations as Independent Precursors to Grokking: A Comparative Study

Abstract

Grokking is a phenomenon in which neural networks trained on small algorithmic tasks achieve near-perfect training accuracy long before test accuracy improves, then suddenly transition to strong generalization. A growing body of concurrent work proposes signals that anticipate this transition before it is visible in test accuracy, drawn from loss-curve spectral oscillations, representation geometry, and loss-landscape curvature. We propose gradient signal-to-noise ratio (SNR) — the ratio of the mean to the standard deviation of the gradient norm over a sliding window — as a leading indicator of grokking onset, and directly test how it relates to an independently proposed precursor signal. Across 15 runs spanning five seeds and three weight decay values on modular addition, gradient SNR decline consistently precedes the test-accuracy jump (15/15 runs, mean lead 2,091 steps, median 1,613, std 821), outperforming a naive training-loss baseline that fails entirely under strong weight decay. We additionally implement a simplified reimplementation of a loss-curve oscillation signal and directly compare its timing against gradient SNR; the proxy did not function as a leading indicator under either configuration tested, which we interpret as a limitation of simplified reimplementation rather than evidence against the underlying phenomenon, and report as an honest, if inconclusive, first attempt at cross-signal comparison in this line of work.

1 Introduction

Neural networks trained on small algorithmic datasets sometimes exhibit “grokking” [7]: training accuracy reaches near-100% within a few thousand steps, while test accuracy remains near chance level for much longer before suddenly rising to near-perfect generalization. A substantial body of early work characterizes this transition only retrospectively. More recently, several independent lines of work propose signals intended to anticipate the transition *before* it is visible in test accuracy: oscillations in the training loss spectrum [10], geometric separation of hidden-layer representations [1], and curvature-based non-commutativity of gradient updates [11]. Each was proposed and validated independently, leaving

open whether these signals detect the same underlying transition or capture distinct information. Our contributions:

1. We propose gradient SNR, a sliding-window ratio of mean to standard deviation of the gradient norm, as a candidate leading indicator of grokking onset, mechanistically distinct from all three signals above.
2. We test this across a sweep of five seeds and three weight decay values, using pre-registered definitions, and show it outperforms a naive training-loss baseline.
3. We implement a simplified reimplementaion of a loss-curve oscillation signal and directly compare its timing against gradient SNR within the same runs — to our knowledge, the first cross-signal comparison in this line of work.

2 Related Work

Grokking was first identified by [7]. Follow-up work proposes retrospective mechanistic and theoretical explanations: internal circuit formation [6], weight norm relative to a critical scale [4], slingshot dynamics in adaptive optimizers [9], and a first-order phase-transition framing via adaptive-kernel feature learning [8]. Separately, [2] show preconditioned gradient descent can mitigate grokking-related delay by reducing spectral bias. None of these propose a signal usable to predict the transition’s timing in advance.

A concurrent cluster does: [10] predict whether a hyperparameter setting will eventually grok at all from early loss-curve spectral oscillations; [1] propose a geometric inter/intra-class distance ratio on held-out representations; [11] propose a curvature-based commutator defect. Each was validated in isolation. Gradient SNR is mechanistically distinct from all three — a time-domain gradient statistic requiring no held-out representations or curvature computation — and to our knowledge no prior work directly compares two such signals within the same runs.

Separately, optimization research links gradient-level statistics to generalization [3, 5], motivating our use of a related quantity here.

3 Method

We study modular addition, $a + b \pmod{p}$, $p = 97$ [7, 10], with a 2-layer, width-128, 4-head decoder-only transformer ($\approx 422,000$ parameters). We train with AdamW ($\text{lr } 10^{-3}$, $\beta = (0.9, 0.98)$, 10-step warmup, minibatch 512), sweeping weight decay $\in \{0.5, 1.0, 2.0\}$ across 5 seeds (15 runs total), 30% training fraction, 15,000 steps on a single T4 GPU.

At each step, we compute the total gradient norm $\|\nabla_t\|$ (L2 norm across all parameters) and define

$$\text{SNR}(t) = \frac{\text{mean}(\|\nabla\|_{t-w:t})}{\text{std}(\|\nabla\|_{t-w:t}) + \epsilon}, \quad w = 50, \epsilon = 10^{-8}. \quad (1)$$

Grokking onset is the first step test accuracy $\geq 90\%$; the **SNR inflection step** is the first step after SNR’s peak at which it falls below 50% of that peak. Both are fixed prior to the sweep.

For the cross-signal comparison, we detrend training loss with a 20-step moving average, then compute Hjorth mobility of the residual over the same window $w = 50$:

$$\text{Mobility}(t) = \frac{\text{std}(\Delta r_{t-w:t})}{\text{std}(r_{t-w:t}) + \epsilon}, \quad (2)$$

and define the **spectral step** using the identical peak-then-decline criterion, applied both across the full run and restricted to the first 1,500 steps.

4 Results

Aggregate results. Across all 15 runs, SNR inflection preceded onset in every case (15/15, mean gap 2,091 steps, median 1,613, std 821; Figure 1). The gap shrinks monotonically with weight decay (mean $\approx 3,108$ at $\text{wd}=0.5$, $\approx 1,612$ at $\text{wd}=1.0$, $\approx 1,552$ at $\text{wd}=2.0$), consistent with weight decay accelerating both signal and transition together. Full per-run results are in Appendix A.

Baseline comparison. A naive baseline (first step training loss < 0.05) leads onset in only 10/15 runs, failing in every run at weight decay 2.0 (negative gap in all 5). Its correlation with onset is strongly negative ($r = -0.816$), moving the wrong direction as weight decay increases, versus $r = 0.620$ for gradient SNR — evidence SNR captures information specific to the generalization transition rather than memorization speed (full comparison, Appendix A).

Cross-signal comparison. Computing the spectral step across the full run, the proxy fired *after* onset in every run (0/15 positive gap, mean $-3,476$ steps), with weak correlation to both onset ($r = 0.131$) and SNR inflection ($r = 0.103$). Restricted to the first 1,500 steps, the proxy’s peak-then-decline pattern did not complete in any run. We interpret this as a limitation of our simplified reimplement — which does not reproduce the fast, early-training oscillatory signature of the original method — rather than evidence the two underlying phenomena are unrelated (Appendix A, Figure 4).

Threshold robustness. Repeating the sweep with the SNR inflection threshold at 30% and 70% of peak, the gap remained positive in all 15 runs at every threshold (correlation with onset: $r = 0.845, 0.620, 0.556$ respectively), confirming the central finding does not depend on the specific 50% threshold chosen.

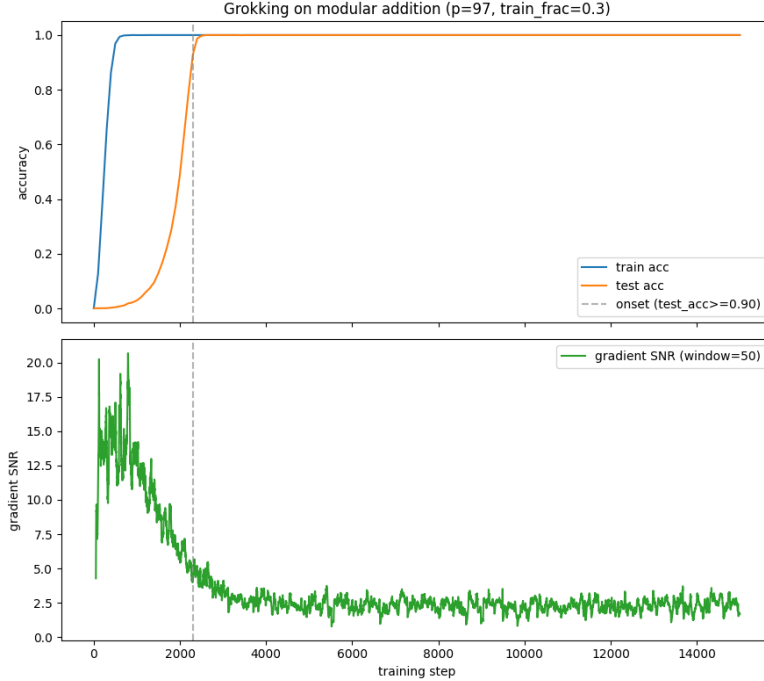


Figure 1: Accuracy and gradient SNR for a representative run (seed 0, wd=1.0).

5 Discussion

The high-SNR phase early in training plausibly reflects the network fitting training data directly, producing large, consistent gradients; as this saturates, gradients become smaller and noisier, coinciding with representational reorganization toward generalization. This is consistent with, though not a direct test of, circuit-formation [6], weight-norm [4], slingshot [9], and phase-transition [8] accounts of grokking. The baseline comparison strengthens this reading: SNR remained reliable across all weight decay values where the naive baseline broke down, suggesting SNR tracks the approach of the generalizing solution rather than memorization speed. Our cross-signal comparison did not yield evidence of agreement between gradient SNR and loss-curve oscillations; we believe this reflects the difficulty of faithfully reproducing a spectral method with a simplified time-domain proxy, rather than an absence of relationship between the two signal types, and leave a rigorous comparison using each method’s original implementation to future work.

Gradient SNR requires no additional forward/backward pass or held-out data beyond training itself — a single scalar per step — in contrast to [1]’s representation extraction or [11]’s additional gradient computations, making it inexpensive for routine monitoring in support of early-stopping or checkpoint-

selection decisions.

Limitations. We study a single task (modular addition, $p = 97$) and architecture; [7] show generalization behavior varies substantially across operations, and [11] find their own signal’s behavior varies across task families, so cross-task generalization of any single precursor signal should not be assumed. Our sweep varies only weight decay and seed; 15 runs is a strong sample for the 15/15 result but modest for the correlation coefficients reported. Our results are correlational: they rule out one naive alternative but do not constitute mechanistic proof that SNR decline causes grokking.

6 Conclusion

Gradient SNR, a simple training-time gradient statistic, reliably precedes grokking onset across 15 runs and outperforms a naive baseline, while our attempt at directly comparing it against an independently motivated loss-curve-oscillation signal was inconclusive — an honest first step toward reconciling a growing, concurrent cluster of grokking precursor-signal proposals. Future work should test generalization across tasks and architectures, and pursue a faithful (rather than simplified) cross-signal comparison.

References

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A Full Sweep Results

Seed	Weight Decay	Onset Step	Inflection Step	Gap	Final Test Acc.
0	0.5	3600	1126	2474	1.000
1	0.5	3800	546	3254	1.000
2	0.5	4200	525	3675	1.000
3	0.5	4100	1258	2842	1.000
4	0.5	4900	1606	3294	0.9998
0	1.0	2300	1205	1095	1.000
1	1.0	2400	852	1548	0.9998
2	1.0	2700	1087	1613	1.000
3	1.0	2700	1153	1547	1.000
4	1.0	3400	1143	2257	0.9998
0	2.0	1700	243	1457	1.000
1	2.0	1700	206	1494	1.000
2	2.0	1900	387	1513	1.000
3	2.0	2100	233	1867	1.000
4	2.0	2100	670	1430	1.000

Table 1: Full sweep results across 5 seeds $\times$ 3 weight decay values (15 runs).

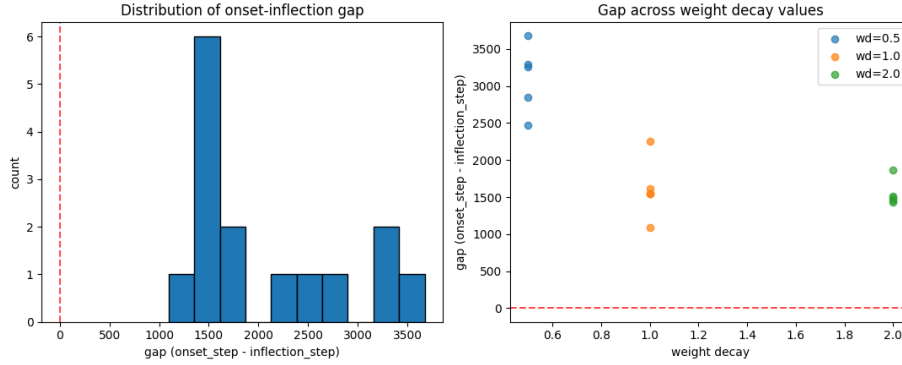


Figure 2: Left: distribution of the onset-inflection gap across all 15 runs. Right: gap as a function of weight decay.

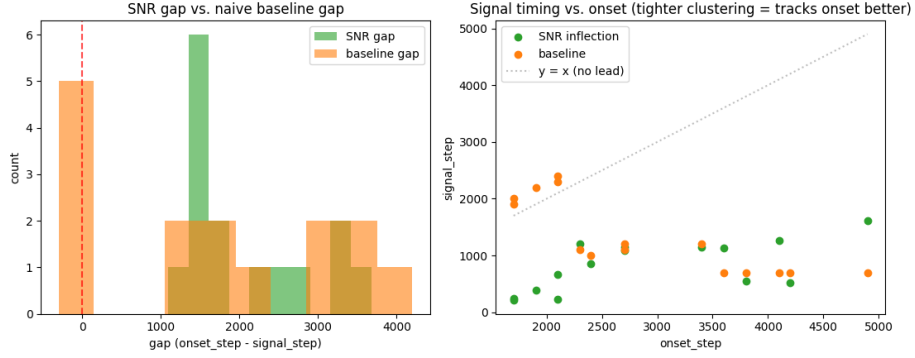


Figure 3: Left: distribution of the onset-signal gap for gradient SNR versus the naive baseline. Right: signal timing versus onset step for both signals.

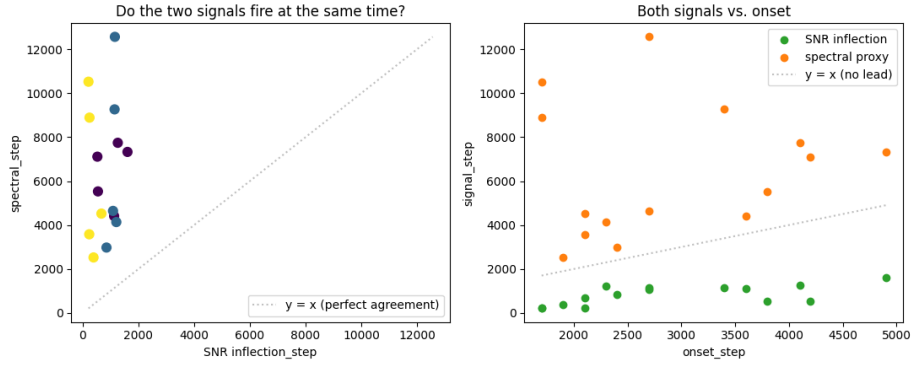


Figure 4: Left: SNR inflection step versus spectral step (full-run configuration); points do not cluster near $y = x$. Right: both signals' timing versus onset step, alongside gradient SNR's tight relationship with onset for comparison.